

2.38 $C_1 = \text{Two-headed coin} \rightarrow P(C_1) = \frac{1}{3}$

$C_2 = \text{Two-tailed coin} \rightarrow P(C_2) = \frac{1}{3}$

$C_3 = \text{Fair coin} \rightarrow P(C_3) = \frac{1}{3}$

a) $P(H_1 | C_1) = 1$ $P(H_1 | C_2) = 0$ $P(H_1 | C_3) = \frac{1}{2}$

$$P(H_1) = P(H_1 | C_1)P(C_1) + P(H_1 | C_2)P(C_2) + P(H_1 | C_3)P(C_3)$$

$$= (1 \times \frac{1}{3}) + 0 + (\frac{1}{2} \times \frac{1}{3}) = \frac{1}{2}$$

$$P(C_1 | H_1) = \frac{1 \times \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

b) $P(C_1 | H_1) = \frac{2}{3}$ $P(C_2 | H_1) = 0$ $P(C_3 | H_1) = \frac{1}{3}$

$$P(H_2 | H_1) = P(H_2 | C_1)P(C_1 | H_1) + P(H_2 | C_3)P(C_3 | H_1)$$

$$= (1 \times \frac{2}{3}) + (\frac{1}{2} \times \frac{1}{3}) = \frac{5}{6}$$

c) $P(H_1 \cap H_2 | C_1) = 1$ $P(H_1 \cap H_2 | C_2) = 0$ $P(H_1 \cap H_2 | C_3) = \frac{1}{4}$

$$P(H_1 \cap H_2) = (1 \times \frac{1}{3}) + (0 \times \frac{1}{3}) + (\frac{1}{4} \times \frac{1}{3}) = \frac{5}{12}$$

$$P(C_1 | H_1 \cap H_2) = \frac{P(H_1 \cap H_2 | C_1)P(C_1)}{P(H_1 \cap H_2)} = \frac{1 \times \frac{1}{3}}{\frac{5}{12}} = \frac{4}{5}$$

2.41

Success = rolling 6 ≥ 1 time out of 4 tries

$$X \sim \text{Binom}(n, p)$$

$$P(X = k) = \binom{n}{k} \cdot p^k (1-p)^{n-k}$$

$$P(X \geq 1) = 1 - P(X = 0)$$

$$= 1 - \text{dbinom}(0, \text{size} = 4, p = 1/6)$$
$$= 0.5177469$$

For 2 dice $= X \sim \text{Binom}(24, 1/36)$

$$= 1 - \text{dbinom}(0, 24, p = 1/36)$$
$$= 0.4914039$$

2.42 114 or more attempts needed, 99 or fewer in first 113 attempts. Needs to miss 14 or more before making 100.

Binomial: $P(\text{Binom}(113, 0.92) \leq 99) \approx 0.0675, 6.7\%$

r code: `pbinom(99, 113, 0.92)`

Ex 2.49) a) $p = \frac{1}{4}$

$X = \text{number correct} \rightarrow X \sim \text{Binom}(20, \frac{1}{4})$

Need at least 12 correct

$$P(X \geq 12) = \sum_{x=12}^{20} \binom{20}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{20-x} \approx 0.00094$$

b) If one choice eliminated, $p = \frac{1}{3}$

$X \sim \text{Binomial}(20, \frac{1}{3})$

$$P(X \geq 12) = \sum_{x=12}^{20} \binom{20}{x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{20-x} \approx 0.013$$

c) P of getting one question correct

$$P = (0.50)(1) + (0.40)\left(\frac{1}{3}\right) + (0.10)\left(\frac{1}{4}\right)$$

$$P = 0.6583$$

$\rightarrow X \sim \text{Binom}(20, 0.6583)$

$$P(X \geq 12) = \sum_{x=12}^{20} \binom{20}{x} (0.6583)^x (0.3417)^{20-x} \approx 0.787$$

2.52 1. Hypothesis

$H_0: \pi = 0.25 \rightarrow$ Spinner is fair

$H_a: \pi \neq 0.25$

2. $X = \#$ of blues, observed value $x = 8$. With H_0 ,
 $X \sim \text{Binom}(50, 0.25)$, expected count would be 12.5.

3. Add up all probability outcomes that are no more
likely than observed one.

$$P(X=8) \approx 0.0463$$

$$\text{Lower T: } P(X \leq 8) \approx 0.0916$$

$$\text{Upper T: } \text{outcome} \leq 0.0463 = X \geq 17.$$

$$\text{Total: } p\text{-value} \approx \boxed{0.190}$$

p is higher than 0.05, therefore we fail to reject H_0 .

r code: `prob <- dbinom(0:50, 50, 0.25)`

`sum(prob[prob <= dbinom(8, 50, 0.25)])`

`binom.test(8, 50, p=0.25)`

2.57 a) $H_0: \pi = 0.5$

$H_a: \pi \neq 0.5$

$p < 0.05$ when $x \leq 8$ and $x \geq 115$

$X \sim \text{Binom}(200, 0.55)$

$$p(X \leq 8) + p(X \geq 115) = 0.262$$

$$p_{\text{binom}}(8, 200, 0.55) + 1 - p_{\text{binom}}(114, 200, 0.55)$$

the probability of getting a p -value less than 0.05
is 0.262.

$$b) \text{SD} = \sqrt{400 \cdot 0.5 \cdot 0.5} = 10$$

$$200 - 1.96 \times 10 = 180.4$$

$$200 + 1.96 \times 10 = 219.6$$

$$p_{\text{binom}}(179, 400, 0.55) + 1 - p_{\text{binom}}(220, 400, 0.55) \\ = 0.481$$

(c) If we execute the command

`1 - sum(dbinom(qbinom(0.025, n, 0.5) : qbinom(0.975, n, 0.5), n, 0.55))`

with $n = 1062$, the result is 0.898. With $n = 1063$, the answer is 0.904. So, 1063 flips are needed.

2.70 a) $X \sim \text{DUnif}(10)$

$$f(x; n) = \begin{cases} \frac{1}{10} & \text{if } x=1, 2, \dots, 10 \\ 0 & \text{otherwise} \end{cases}$$

i. $P(X=3) = \frac{1}{10}$

ii. $P(X=12) = 0$

iii. $P(X \leq 3) = \sum_{i=1}^3 P(X=i) = \frac{1}{10} + \frac{1}{10} + \frac{1}{10} = \frac{3}{10}$

b) $X \sim \text{Unif}(n)$

$$E(X) = \sum_{x=0}^n x f(x) = \sum_{x=0}^n x \cdot \frac{1}{n} = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 = \sum x^2 f(x) - \left(\frac{n+1}{2}\right)^2 = \frac{1}{n} \sum x^2 - \frac{(n+1)^2}{4} \\ &= \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = (n+1) \left(\frac{2n+1}{6} - \frac{n+1}{4} \right) = \frac{n^2-1}{12} \end{aligned}$$

2.81

$$E(X) = E(Y) = 10 \quad \text{Var}(X) = \text{Var}(Y) = 2$$

(a) $E(X+Y) = E(X) + E(Y) = 10 + 10 = 20$

$\text{Var}(X+Y)$, independent $= \text{Var}(X) + \text{Var}(Y) = 2 + 2 = 4$

(b) $E(2X) = 2E(X) = 2(10) = 20$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$= 2^2 \text{Var}(X) = 4 \cdot 2 = 8$$

(c) for $X+Y$, since they are independent, so Mean will remain ~~but Variance~~ would

Since r.v. can sometimes cancel each other, making var does not always increase in the same direction.

but for $X+X$, same r.v. twice. so deviation are doubled.

#2.88

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$Y = X^2 \rightarrow XY = X^3$$

$$\begin{aligned} E(X) &= (-1)(0.25) + (0)(0.5) + (1)(0.25) \\ &= -0.25 + 0 + 0.25 = 0 \end{aligned}$$

$$\begin{aligned} E(Y) &= E(X^2) = (-1)^2(0.25) + (0)^2(0.5) + (1)^2(0.25) \\ &= 0.25 + 0 + 0.25 = 0.5 \end{aligned}$$

$$E(XY) = E(X^3) = 0$$

$$\text{Cov}(X, Y) = 0 - 0(0.5) = 0$$

X and Y are not independent $P(X=0 \text{ and } Y=1) = 0$
 $P(X=0) \cdot P(Y=1) = 0.5 \cdot 0.5 = 0.25$